Solution Requirements

Date	02 July 2026
Team ID	SWTID-2026-8076
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*The system shall allow applicants and administrators to securely log in using email/mobile number and password with OTP verification for sensitive actions.*	High
2	Authorization levels	*The system shall provide role-based access where Applicants can submit and track applications, while Administrators can verify applications, manage users, and view prediction reports.*	High

3	External interfaces	*The system shall integrate with Credit Bureau APIs, Aadhaar/PAN verification services, bank databases, and email/SMS notification services for applicant verification and communication.*	High
4	Transactions processing	*The system shall receive applicant information, validate data, preprocess financial details, execute the machine learning prediction model, and generate approval or rejection results in real time.*	High
5	Reporting	*The system shall generate applicant status reports, approval/rejection reports, prediction accuracy reports, dashboards, and downloadable PDF/Excel reports for administrators.*	Medium
6	Business rules, etc.	*Credit card approval shall be determined using predefined eligibility rules such as minimum income, credit score, debt-to-income ratio, employment status, and ML prediction score.*	High
7	Compliance to laws or regulations	*The system shall comply with data privacy regulations, secure storage of personal information, banking security standards, and obtain user consent before processing personal data.*	High

8	*Other*	*The system shall send automatic email/SMS notifications regarding application submission, verification progress, approval, or rejection status.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	*The system shall process applicant data and display the credit card approval prediction quickly with minimal response time.*	*Prediction result displayed within 3 seconds for 95% of requests.*
2	Scalability	*The system shall efficiently support an increasing number of users and applications without performance degradation*	*Supports at least 10,000 concurrent users and processes 50,000+ applications per day.*
3	Security & Data Privacy	*The system shall securely protect applicant data using encryption, authentication, and secure communication protocols.*	*AES-256 encryption for stored data, HTTPS/TLS for data transmission, Role-Based Access Control (RBAC) enabled.*
4	Reliability & Availability	*The system shall remain operational with minimal downtime and provide reliable prediction results.*	*(e.g., 99.9% uptime per month) 99.9% monthly uptime, automatic backup every 24 hours, recovery within 15 minutes after failure.*

5	Usability & Accessibility	*The system shall provide an intuitive, responsive, and user-friendly interface accessible across multiple devices and browsers.*	*Responsive design, compatible with major browsers, and follows WCAG 2.1 AA accessibility guidelines.*
6	*Other*	*The system shall be modular, easy to maintain, support future machine learning model updates, and be deployable on different platforms.*	Modular architecture, deployment on Windows, Linux, and Cloud platforms, and ML model updates without affecting existing functionality.